UEI733: EMBEDDED SYSTEM DESIGN

L	T	P	Cr
3	0	2	4.0

Introduction: Introduction to Embedded Systems, Its Architecture and system Model, Microprocessors &Microcontrollers, Introduction to the ARM Processor architecture, Embedded Hardware Building Block. Embedded System Attributes.

ARM Microprocessor Architecture: ARM Core M4, Architecture, Power architecture, Low power modes, Registers, Memory organization and system, Input & Output port, ARM Cortex Microcontroller Software Interface Standard (CMSIS), CMSIS library for DSP operations, CMSIS for artificial intelligence, Exceptions and Interrupts / NVIC / EXTI & Inter-Module Signaling, Communications protocols: SPI, RS232, I2C, Built-in ADC and DAC.

EMBEDDED PROGRAMMING: C/C++ language programming concepts, Declarations and Expressions, Arrays, Decision and Control Statements, interrupts and exception programming, Interfacing with ADC/DAC and LCD/Touchscreen programming. Development tools and Programming: Compiler, Assembler and Debugger, Hardware and Software development tools, Code warrior-IDE or STM32CubeMX IDE/ KEIL/Mbed etc., General-Purpose Input Output (GPIO) and the STM Hardware Abstraction Layer (HAL).

Real-time Operating Systems (RTOS) : Basic concepts of RTOS and its types, Task, process & threads, interrupt routines in RTOS, Multiprocessing and Multitasking, scheduling, Concurrency, Introduction to open source RTOS like FreeRTOS/MQX.

IoT in an embedded system: Introduction to WiFI, BLE and Zigbee wireless technologies, IoT protocols, Creating a mini internet of things (IoT) system, IoT clouds like Azure/ AWS etc.

Laboratory Work:

Programming of microcontroller with Integrated development environment (IDE), Interfacing with switches, sensor modules, with their Programming, LCD/ Touch screen interfacing, Interfacing of Temperature Sensors with I2C/SPI bus, Interfacing to Motors, relays and other actuators, Making the embedded system compatible with IoT.

The students can implement the experiments on STEM32 or FRDM-KL25 boards

Case study: Measuring Current, Power, and Energy on the FRDM-KL25Z board.

Course Learning Outcomes (CLO):

Student will be able to:

1. Exhibit the knowledge of ARM Microprocessor Architecture
2. Exhibit the knowledge of programming of C/C++ language for embedded applications
3. Elucidate the knowledge of interfacing of sensors and actuators.
4. Exhibit the knowledge of real-time operating systems

Text Books:

1. *Alexander G. Dean , Embedded Systems Fundamentals with Arm Cortex M Based Microcontrollers: A Practical Approach, ARM Education media, 2017.*
2. *Haung, H.W., The HCS12 / 9S12: An Introduction to Software and Hardware Interfacing, Delmar Learning (2007).*

Reference Books:

1. *Fredrick, M.C., Assembly and C programming for HCS12 Microcontrollers, Oxford University Press (2005).*
2. *Ray, A.K., Advance Microprocessors and Peripherals – Architecture, Programming and Interfacing, Tata McGraw-Hill (2007)*

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST+EST	70
2	Sessional (May include Assignments//Quizzes/Lab Evaluations)	30